ILINA BHAYA-GROSSMAN

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EDUCATION

University of California, San Francisco *2020 – 2025*
Ph.D., Cognitive Neuroscience

UCSF–UC Berkeley Joint Graduate Program
Dissertation: "Cross-language speech computations in the human temporal cortex"
Advisor: Prof. Edward F. Chang
Committee: Prof. Frederic E. Theunissen, Prof. Keith Johnson, Prof. Matthew K. Leonard

University of California, Berkeley *2014 – 2018*
B.A. in Cognitive Science and Computer Science (Magna cum laude)

Highest Honors in Cognitive Science; GPA 3.92

RESEARCH APPOINTMENTS

Stanford Gwilliams Lab *2026 – Present*
Postdoctoral Scholar

Research on the neural correlates of nonspeaking autism using human intracranial EEG and optically-pumped-magnetometer magnetoencephalography (OPM-MEG). *Advised by Prof. Laura Gwilliams*

UCSF Chang Lab *2020 – 2025*
Graduate Student

Doctoral research on the neural correlates of language experience in the temporal lobe using human electrocorticography (ECoG). *Advised by Prof. Edward F. Chang*

UC Berkeley Kavli Center for Ethics, Science, and the Public *2024 – 2025*
Graduate Fellow

Examined the ethical and social implications of emerging neurotechnologies and science policy. *Advised by Dr. Lea Witkowsky and Prof. Jodi Halpern*

Basque Center for Cognition, Brain, and Language *2023*
Visiting Scholar

Research on the mechanisms of cross-lingual speech comprehension in EARSHOT, a computational model of spoken word recognition. *Advised by Prof. Jim Magnuson*

UC Berkeley Knight Lab *2016 – 2018*
Research Assistant

Research on signal coherence of scalp EEG across participants (hyperscanning) during a non-verbal communication paradigm. *Advised by Prof. Robert Knight and Dr. Arjen Stolk*

Stanford University Huang Lab *2012 – 2014*
Research Assistant

Developed CUDA GPU processing code for automated bacterial cell annotation software. *Advised by Prof. KC Huang*

HONORS & AWARDS

SNL Dissertation Award, 2026

National Science Foundation Graduate Research Fellowship (GRFP), 2020–2025

1st Place, UCSF Grad Slam, 2024

1st Place, Cognitive Science Society Mind Challenge, 2023

H2H8 Explorer Award, 2023

UCSF Discovery Fellow, 2022

Robert J. Glushko Prize for Undergraduate Research, UC Berkeley, 2018

Highest Honors in Cognitive Science, UC Berkeley, 2018

Travel Awards:

Professional Development Award, DANA Foundation (2025)
Trainee Scholarship, Society for the Neurobiology of Language (2024)
Travel Award, UC Berkeley Graduate Division (2024)
Travel Award, Advances and Perspectives in Auditory Neuroscience (2023)
Travel Award, Society for the Neurobiology of Language (2023)
Conference Travel Award, Associated Students of the Graduate Division (2022)
Training and Professional Development Award, Society for Neuroscience (2021)
Travel Award, Hong Kong University of Science and Technology Hackathon (2017)
Travel Award, Tapia Conference for Diversity in Computing (2015)

PUBLICATIONS

† *denotes publications from doctoral dissertation research; * denotes equal contribution.*

Peer-Reviewed Publications

† **Bhaya-Grossman I.**, Leonard M., Zhang Y., Gwilliams L., Chang E.F. (2025). "Shared and language-specific phonological encoding in the human temporal lobe." *Nature*.

† Zhang Y.*, Leonard M.*, **Bhaya-Grossman I.**, Gwilliams L., Chang E.F. (2025). "Dynamics of auditory word form encoding in the human temporal cortex." *Neuron*.

Gwilliams L.*, **Bhaya-Grossman I.***, Zhang Y.*, Scott T.*, Harper S.*, et al. (2024). "Computational Architecture of Speech Comprehension in the Human Brain." *Annual Review of Linguistics*.

Silva A., Liu J., Metzger S., **Bhaya-Grossman I.**, et al. (2023). "A Bilingual Speech Neuroprosthesis." *Nature Biomedical Engineering*.

† Oganian Y.*, **Bhaya-Grossman I.***, Chang E.F. (2023). "Vowel and formant representation in the human auditory speech cortex." *Neuron*.

† **Bhaya-Grossman I.**, Chang E.F. (2021). "Speech Computations in the Human Superior Temporal Gyrus." *Annual Review of Psychology*, Vol. 73.

Llorens A.*, Tzovara A.*, **Bhaya-Grossman I.**, Bidet-Caulet A., Chang W., et al. (2021). "Gender bias in academia: A lifetime problem that needs solutions." *Neuron*.

Ursell T.S., Lee T.K., Shiomi D., Shi H., Tropini C., Monds R.D., Colavin A., Billings G., **Bhaya-Grossman I.**, et al. (2017). "Rapid, precise quantification of bacterial cellular dimensions across a genomic-scale knockout library." *BMC Biology.*

Under Review

Silva A., Leonard M., Hullett P., Levy D., Wang C., Xu D., **Bhaya-Grossman I.**, Chang E.F. "A neural code for generative speech production in human Wernicke's area."

In Preparation

† **Bhaya-Grossman I.**, Oganian Y., Grabowski E., Chang E.F. "Emergence of lexical stress encoding in the human temporal cortex."

Sankaran N., **Bhaya-Grossman I.**, Krakowski A., Bishop L., Greenwald E., et al. "Can public dialogue shape neuroscientists' views on the ethics of neurotechnologies? Evidence from a field experiment."

INVITED TALKS

Cognitive Science Colloquium, UC Berkeley (2026)
Phonetics, Phonology and Psycholinguistics Forum (Phorum), UC Berkeley (2026)
Language and Computation in Neural Systems, Max Planck Institute for Psycholinguistics (2026)
Woolgar Lab, MRC Cognition and Brain Sciences Unit, Cambridge University (2026)
Guest Lecture, Psychology of Language, Mills College at Northeastern (2026)
Language, Intelligence, and Thought Lab, Georgia Tech (2026)
Computational Neuroscience of Audition Lab, University of Rochester (2026)
ARCS (Achievement Rewards for College Scientists) Symposium (2025)
Electronic Auditory Research Seminars (EARS) (2024)
Society for the Neurobiology of Language (SNL) Annual Meeting (2024)
Bay Area Language Processing Workshop, Stanford University (2024)
UCSF Neurosciences Research in Progress Talks (2024)
Weill Institute for Neuroscience Seminar Series (WINSS), UCSF (2024)
McDonald Lab Journal Club, UC San Diego (2024)
Advances and Perspectives in Auditory Neuroscience (APAN) (2023)
UC Irvine Center for Hearing Research (2023)
University of Tübingen, Center for Integrated Neuroscience (2023)
Basque Center for Cognition, Brain, and Language (2023)

CONFERENCE POSTERS

Kries J., Anand R., **Bhaya-Grossman I.**, Peony O., Lucas M., Yamada L., Fogarty A., Nuyujukian P., Deisseroth K., Gwilliams L., Baumer F. "Neural encoding of speech in children with epilepsy: an intracranial study." *Maternal Health and Child Research Institute Conference (2026).*

Bhaya-Grossman I., Oganian Y., Grabowski E., Chang E.F. "Emergence of lexical stress encoding in the human temporal cortex." *Gordon Research Conference (2026); Gordon Research Seminar (2026); Society for Neurobiology of Language (2026).*

Anand R., Kries J., **Bhaya-Grossman I.**, Baumer F., Gwilliams L. "Investigating the Effect of Epileptiform Activity on Speech Envelope Tracking in Children with Epilepsy." *Stanford BioX Research Conference (2026).*

Sankaran N., **Bhaya-Grossman I.**, Krakowski A., Bishop L., Greenwald E., Witkowsky L. "Can public dialogue shape neuroscientists' views on the ethics of neurotechnologies? Evidence from a field experiment." *International Neuroethics Conference (2026).*

Wang C.*, **Bhaya-Grossman I.***, Chang E.F. "Language impairment profile of a polyglot following neurosurgical resection: a case report." *Academy of Aphasia (2025).*

Bhaya-Grossman I., Oganian Y., Grabowski E., Chang E.F. "Neural Encoding of the Rhythm of Words in Human Speech Cortex." *Sensation and Action (2025); Cognitive Computational Neuroscience [CCN] (2025).*

Bhaya-Grossman I., Oganian Y., Grabowski E., Chang E.F. "Encoding of Relative Speech Intensity in the Human Temporal Cortex." *Society for Neuroscience [SfN] (2024); Advances and Perspectives in Auditory Neuroscience [APAN] (2024).*

Bhaya-Grossman I., Leonard M., Zhang Y., Gwilliams L., Johnson K., Chang E.F. "Language-familiarity Dependent Encoding of Speech in Human Temporal Lobe." *ASCONA Neural Circuits Conference (2024); Society for the Neurobiology of Language [SNL] (2023).*

Zhang Y., Gwilliams L., **Bhaya-Grossman I.**, Leonard M., Chang E.F. "Segmenting Words from Continuous Speech in the Human Temporal Cortex." *SNL (2023); APAN (2023).*

Bhaya-Grossman I., Oganian Y., Johnson K., Chang E.F. "Vowel Encoding In The Human Superior Temporal Gyrus." *SfN (2021); SNL (2020); APAN (2020).*

SERVICE & OUTREACH

Ad hoc Reviewer: *Nature Communications*; *Scientific Reports* (Springer Nature); *Communications Psychology*; *Cognitive Computational Neuroscience* (CCN); *Hearing Research*; *Journal of Phonetics*

Organizer, Bay Area Language Processing Workshop (2024)

Designer, Illustrator, and Contributor, *Berkeley Science Review* (2024)

TEACHING

University of California, San Francisco *2020 – Present*

Machine Learning for Medical Imaging
Teaching Assistant, Department of Bioinformatics (2025)

Speech and Language Processing in the Brain
Course Designer and Instructor, Department of Neurosurgery (2024)

Chang Lab Python for Neuroscience Tutorial
ENVISION Facilitator, Department of Neurosurgery (2020 – 2024)

University of California, Davis *2019*

sEEG/ECoG Fieldtrip Bootcamp
Programming Tutor, Department of Neurosurgery

University of California, Berkeley *2015 – 2017*

Data Structures and Interpretation of Computer Programs
Tutor, Department of Computer Science (2015 – 2016)

Python for Social Scientists
Teaching Assistant, Data-Lab (2016 – 2017)

MENTORING

Allison Lievano Gomez, Undergraduate RA (2026)
Riya Anand, Undergraduate RA (2026)
Shrujana Kannam, Summer Intern (2025)
Reilly Sheehan, Summer Intern (2024)
Ashley Qin, Rotation Student (2024)
Marcos Lobato Scharfhausen, Summer Intern (2022)
Carolina Varona Arguelles, Summer Intern (2022)

PROFESSIONAL MEMBERSHIPS

Society for the Neurobiology of Language (2019 – Present)

Society for Neuroscience (2020 – Present)

Phi Beta Kappa Honor Society (2017 – Present)

Upsilon Pi Epsilon, Computer Science Honor Society (2016 – 2018)

LANGUAGES & SKILLS

Languages: English (native), Chinese (advanced)
Programming: Python, MATLAB, LaTeX, C, C++, SQL, Go, Angular, HTML, CSS